ORDER CLOSURE, ORDER ADHERENCE AND FATOU NORMS

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Abstract. We record two results in the theory of vector and Banach lattices related to order adherence. First, we give a negative answer to Gao and Leung's question on whether the *uo*-adherence of sublattices must be order closed. Second, we present a self-contained counterexample showing that a Banach lattice with a weakly Fatou norm need not admit any equivalent lattice norm with the Fatou property.

1. Introduction

Recall that a net (x_α) in a vector lattice X *order converges* to $x \in X$, denoted $x_\alpha \xrightarrow{o} x$, if there exists a net $y_\beta \downarrow 0$ (decreasing with infimum 0) in X such that for all β there exists α_0 with

$$|x_\alpha - x| \leq y_\beta,$$

for any $\alpha \geq \alpha_0$. The net (x_α) *uo-converges* (unbounded order converges) to $x \in X$, denoted $x_\alpha \xrightarrow{uo} x$, if

$$|x_\alpha - x| \wedge u \xrightarrow{o} 0$$

for all $u \in X_+$. In spaces of measurable functions, it is well-known that a sequence *uo*-converges to zero if and only if it converges to zero pointwise almost everywhere (i.e. up to a set of zero measure). On the other hand, Bilokopytov and Troitsky [3] have recently shown that in spaces of continuous functions $C(K)$, a sequence *uo*-converges to zero if and only if it converges to zero pointwise on a comeagre set. For further information on order and *uo*-convergence we refer the reader to [7, 9, 12, 13].

Given a notion of convergence $c \in \{o, uo\}$, a subset Y of a vector lattice X is *c-closed* if $\overline{Y}^c = Y$. The *c-closure* of Y in X is the smallest c-closed subset of X containing Y. We will denote the *c-adherence* of Y in X by $\overline{Y}^c := \{x \in X : \exists\ (y_\alpha) \subseteq Y \text{ with } y_\alpha \xrightarrow{c} x\}$. Our aim here will be to exploit the fact that unlike with topological closures, $\overline{Y}^c$ need not be c-closed and that an arbitrarily large number of iterations of c-adherence might be needed to reach the c-closure of a set.

With the motivation of providing a general approach to a fundamental result in financial economics concerning the spanning power of options written on a financial asset, Gao and Leung investigated the question of identifying the order closure of a given sublattice [8]. Let X be a Banach lattice. By passing to a summable subsequence, it is easy to see that norm convergent sequences have order convergent subsequences. Hence, if X is order continuous – meaning that order convergence implies norm convergence in X – then $\overline{Y}^o$ will be order closed for every subset $Y \subseteq X$. Remarkably, Gao and Leung were able to establish the converse to this statement, even after restricting their attention to sublattices. More specifically, they proved the following theorem.

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Theorem 1.1 ([8], Theorem 2.7)**.** *Let X be a σ-order complete Banach lattice. The following statements are equivalent.*

(1) The order adherence of every sublattice of X is order closed.
(2) The order and uo-adherence of every sublattice agree.
(3) X is order continuous.

In addition to Theorem 1.1, they were able to prove in [8, Lemma 2.1] that if Y is a sublattice of a vector lattice X, then we have the inclusions

$$\overline{Y}^{o} \subseteq \overline{Y}^{uo} \subseteq \overline{\overline{Y}^{o}}^{o}.$$

Moreover, if $\overline{Y}^{uo}$ is order closed, then it coincides with the order closure of Y, and $\overline{Y}^{uo} = \overline{\overline{Y}^{o}}^{o}$. In this case, it follows that $\overline{\overline{\overline{Y}^{o}}^{o}}^{o} = \overline{\overline{Y}^{o}}^{o}$. Together with statements (i) and (ii) of Theorem 1.1, these observations motivate the question of whether the *uo*-adherence of a sublattice Y is always order closed. Indeed, this question was explicitly stated in Problem 2.5 of their paper.

Question 1.2 ([8], Problem 2.5)**.** Is $\overline{Y}^{uo}$ order closed for every sublattice of a vector lattice X?

In Section 2, we give a strong negative resolution to this problem by showing that there is no universal bound on the number of iterated order adherences one must take before reaching an order closed sublattice. This is done by translating into the Banach lattice framework the analogous construction for Boolean algebras first due to Gaifman [6] and Hales [10] and later simplified by Solovay [11]. The remainder of this section contains a discussion on the size of the order adherence of a solid set S in terms of the size of a generating set of S. In particular, we will show in Theorem 2.6 that solid sets may also need an arbitrarily large number of order adherence iterations to reach their order closure.

The second purpose of the paper is to present a self-contained solution to a problem originally posed by D. Fremlin [5] on weakly Fatou norms and resolved by M. Elliot in [4]. Recall that a lattice norm $\|\cdot\|$ on a vector lattice X has the *Fatou property* if

$$\|x\| = \sup_{\alpha} \|x_\alpha\|$$

whenever $0 \leqslant x_\alpha \uparrow x$ in X. It is *weakly Fatou* if there exists $K \geqslant 1$ such that

$$\|x\| \leqslant K \sup_{\alpha} \|x_\alpha\|$$

whenever $0 \leqslant x_\alpha \uparrow x$. Clearly, every equivalent renorming of a Fatou norm is weakly Fatou. Fremlin's question is whether the converse holds:

Question 1.3 ([5], Problem AB)**.** Is every weakly Fatou norm equivalent to a Fatou norm?

We emphasize that the solution to the above problem is known. Indeed, a solution was announced by A. Wickstead and credited to M. Elliott [4] at the Positivity IX conference in Edmonton, Alberta, 2017. However, the proof has never been published or publicly released. In Section 3, we present a self-contained construction of a weakly Fatou norm on a Banach lattice which is not equivalent to a Fatou norm.

The connection between Fremlin's problem and order adherence is very natural: If $\|\cdot\|$ is a weakly Fatou norm on X, then there is some $K > 0$ such that the unit ball satisfies

$$\overline{B_{(X,\|\cdot\|)}}^{o} \subset K B_{(X,\|\cdot\|)}.$$

Iterating this process n times, we get that

$$\overline{\overline{B_{(X,\|\cdot\|)}}^{o^{\cdot^{\cdot^{\cdot}}}}}^{o} \subset K^n B_{(X,\|\cdot\|)}.$$

On the other hand, if $\|\cdot\|$ is Fatou we must have $\overline{B_{(X,\|\cdot\|)}}^{o} = B_{(X,\|\cdot\|)}$.

In our approach, we will produce for every natural number n a norm on a space of continuous functions which is weakly Fatou with constant 2, but whose n-th iterated order adherence of the unit ball has an element of norm 2^n. This will force any Fatou norm on the space to have a bad equivalence constant with the original norm. Gluing these spaces together for $n \in \mathbb{N}$ we obtain the claimed construction.

Our motivation for giving an example of a solid set whose order adherence is not order closed was to correct a mistake made in [12], which wrongly assumed that an increasing supremum of increasing supremum could be written as an increasing supremum. We thank E. Bilokopytov and K. Abela for pointing out this mistake to the author. After several months of work, we produced the material in the first two sections of the present paper. Given the above connection to Fremlin's problem, it was natural to ask whether our ideas could also give a simpler solution to this problem. However, constructing the norm so that the solid set whose order adherence keeps expanding is the unit ball is highly non-trivial. To overcome this, we rely heavily on ideas from [4]. The ideas in [4] are ingenious, but the execution is quite complicated and indirect. Our main contribution is to simplify and distill these ideas to produce a Lean-verified and easily digestible proof that we believe will be useful to the community.

1.1. **Lean formalization.** The results presented in this paper have been formalized in Lean 4 using version `v0.1.0` of `banlat`, a Lean library for Banach lattices. This version of the library, in turn, depends on Mathlib version `v4.30.0` `mathlib4`. Each of the results in Section 2 and Section 3 includes a link to the corresponding Lean declaration pointing to a specific line in `OrderClosures`. For completeness we have also included formalization of Solovay's construction from [11], which is fundamental for the proof of Proposition 2.1, as well as the formalization of Theorem 1.1 from [8].

2. The Gao-Leung problem on uo-adherences of sublattices

The next result gives a strong negative answer to the Gao-Leung problem.

Proposition 2.1. [∀ `gao_counterexample`] *For any cardinal κ there exists a compact Hausdorff space K such that $C(K)$ is order complete and has density character at least κ, together with a separable norm closed sublattice $Y \subseteq C(K)$ such that the only order closed sublattice of $C(K)$ containing Y is $C(K)$ itself.*

Remark 2.2. To see that Proposition 2.1 answers the question of Gao and Leung, we include a simple cardinality estimate. Let D be a subset of a vector lattice X of cardinality σ, and let $\overline{D}^o$ denote the order adherence of D in X. We claim that

$$|\overline{D}^o| \leqslant 2^{2^\sigma}.$$

Fix $x \in \overline{D}^o$ and let

$$\mathcal{D}(x) = \{A \subseteq D : x \in \overline{A}^o\} \in \mathscr{P}(D).$$

Since $\mathcal{D}(x)$ is a family of subsets of D, we have $|\mathcal{D}(x)| \leqslant 2^\sigma$. Now suppose $x, y \in \overline{D}^o$ and $x \neq y$. We prove that $\mathcal{D}(x) \neq \mathcal{D}(y)$. This yields

$$|\overline{D}^o| = |\{\mathcal{D}(x) : x \in \overline{D}^o\}| \leqslant |\mathscr{P}(\mathscr{P}(D))| = 2^{2^\sigma}.$$

To prove the claim, let (x_α) be a net in D such that $x_\alpha \xrightarrow{o} x$. Let $y_\alpha \downarrow 0$ be such that $|x_\alpha - x| \leqslant y_\alpha$ eventually. Although it is not strictly necessary, we may assume that (y_α) has the same index set as (x_α) after passing to a subnet; see [1]. Since $x \neq y$, there exists α_1 such that $|x - y| \nleqslant y_{\alpha_1}$. Let

$$A = \{x_\alpha : \alpha \geqslant \alpha_1\}.$$

Then clearly $A \in \mathcal{D}(x)$. We show that $A \notin \mathcal{D}(y)$. Suppose towards a contradiction that there is a net (a_γ) in A with $a_\gamma \xrightarrow{o} y$. Passing to a subnet, choose $z_\gamma \downarrow 0$ with $|a_\gamma - y| \leqslant z_\gamma$ for all γ. Then

$$|x - y| \leqslant |x - a_\gamma| + |y - a_\gamma| \leqslant y_{\alpha_1} + z_\gamma.$$

Passing to the limit gives $|x - y| \leqslant y_{\alpha_1}$, a contradiction. Thus $\mathcal{D}(x) \neq \mathcal{D}(y)$, as claimed.

Proof of Proposition 2.1. We begin with the complete Boolean algebra $\mathbb{B}$ of Solovay [11] with $|\mathbb{B}| \geqslant \kappa$ that contains a countable subset $G \subseteq \mathbb{B}$ with the property that there does not exist a complete Boolean subalgebra $\widetilde{\mathbb{B}}$ satisfying $G \subseteq \widetilde{\mathbb{B}} \subsetneq \mathbb{B}$.

Let K be the Stone space of $\mathbb{B}$. A Boolean algebra is complete if and only if its Stone space is extremally disconnected, which is equivalent to $C(K)$ being order complete. We represent $G \subseteq \mathbb{B} \subseteq C(K)$ in the standard way by indicator functions. Since $|\mathbb{B}| \geqslant \kappa$, the space $C(K)$ has density character at least κ, because the elements of $\mathbb{B}$ are all 1-separated.

Let Y be the closed sublattice of $C(K)$ generated by G and $\mathbf{1}$. Then Y is separable. Since $C(K)$ is order complete, every order closed sublattice of it is order complete and norm closed: norm convergent sequences have subsequences that converge in order. It therefore suffices to show that if X is an order closed and norm closed sublattice of $C(K)$ containing Y, then $X = C(K)$.

Set

$$\widetilde{\mathbb{B}} = \mathbb{B} \cap X.$$

Clearly $G \subseteq \widetilde{\mathbb{B}}$. We claim that $\widetilde{\mathbb{B}}$ is a complete Boolean subalgebra of $\mathbb{B}$. Once this is proved, the defining property of G yields $\widetilde{\mathbb{B}} = \mathbb{B}$. Hence $\mathbb{B} \subseteq X \subseteq C(K)$. Since $\mathbf{1} \in X$ and indicator functions of clopen sets separate points of the Stone space, Stone–Weierstrass gives $X = C(K)$.

It remains to prove that $\widetilde{\mathbb{B}}$ is complete. The Boolean operations are inherited from $\mathbb{B}$, so only completeness requires an argument. Let $A \subseteq \widetilde{\mathbb{B}}$, and let $b = \sup A$ in the complete Boolean algebra $\mathbb{B}$. Viewed inside $C(K)$, the indicator function of b is the order supremum of the family of indicator functions of elements of A. Since those indicator functions belong to the order closed sublattice X, their supremum also belongs to X. Thus $b \in \widetilde{\mathbb{B}}$, and $\widetilde{\mathbb{B}}$ is complete. ☐

In everything that follows (including Section 3) we shall apply order adherence only to solid sets. Note that the order adherence of a solid set is also solid. For simplicity, we will introduce some notation – we leave it as a standard exercise to show that the definition of order adherence below is consistent with the one above.

Definition 2.3. Let X be a vector lattice and let $A \subseteq X$ be solid. The *order adherence* A^o is the solid hull of the set of all $x \in X_+$ for which there exists an upward directed set $B \subseteq A \cap X_+$ with $\sup B = x$ in X.

Inductively, put

$$A^{o(0)} := A, \qquad A^{o(m+1)} := \big(A^{o(m)}\big)^o.$$

Let $\mathrm{solid}(A)$ be the least solid set that contains A. Namely,

$$\mathrm{solid}(A) = \bigcup_{a \in A} [-|a|, |a|].$$

Given a solid set S, let $\mathrm{solidgen}(S)$ be the least cardinality of a set A such that $S = \mathrm{solid}(A)$.

Proposition 2.4. [∀ `exists_solid_large_orderAdherence`] *For any cardinal κ there exists a solid set S with $\mathrm{solidgen}(S) = \omega$ but $\mathrm{solidgen}(S^o) \geqslant \kappa$.*

Proof. Consider the following Banach sublattice X of $\ell_\infty(\Gamma \times \omega)$:

$$X = \{ f \in \ell_\infty(\Gamma \times \omega) : \exists x \in c_0(\Gamma)\ \exists y \in c_0(\mathbb{N}) : |f(\gamma, n)| \leqslant x(\gamma) + y(n) \}.$$

For every n, the characteristic function $\chi_{\Gamma \times \{n\}}$ belongs to X, because

$$\chi_{\Gamma \times \{n\}}(\gamma, n) \leqslant 0(\gamma) + \chi_{\{n\}}(n).$$

Let S be the solid subset of X generated by the countable set $\{\chi_{\Gamma \times \{n\}} : n \in \mathbb{N}\}$. Since S contains all finitely supported sequences, every element of X belongs to the order closure, and even to the order adherence, of S.

We now estimate $\mathrm{solidgen}(X)$. Given $f \in X$, notice that

$$\Gamma_f = \{\gamma \in \Gamma : \chi_{\{\gamma\} \times \mathbb{N}} \leqslant f\}$$

is a finite subset of Γ. This is because $\chi_{\Delta \times \mathbb{N}} \notin X$ if Δ is infinite. Therefore, if $X = \mathrm{solid}(A)$, then we must cover Γ with $|A|$ many finite sets. This implies that $|A| \geqslant |\Gamma|$. ☐

There is, however, a cardinality bound on the size of the order adherence.

Proposition 2.5. [∀ `solid_orderAdherence_cardinality_bound`] *If S is a solid set, then*

$$|S^o| \leqslant 2^{|S|}.$$

Proof. It is enough to check that

$$S^o \subseteq \left\{ x : x^+, x^- \in \left\{ \bigvee A : A \subseteq S^+ \right\} \right\}.$$

For this, it is enough to prove that the right-hand side is order closed. Since taking positive and negative parts are order continuous operations, it suffices to prove that

$$Z = \{ \bigvee A : A \subseteq S^+ \}$$

is order closed. Suppose $z = o\text{-}\lim z_i$ with $z_i \in Z$. In particular, we have an increasing net $y_k \leq z_i$ with $\sup y_k = z$. By taking positive parts, we may suppose that $0 \leqslant y_k$.

It remains to show that each $y_k \in Z$. But $0 \leqslant y_k \leqslant z_i \in Z$ for some i. So if $z_i = \sup A$, then

$$y_k = \sup\{a \wedge y_k : a \in A\}.$$

To justify this, suppose that $w < y_k$ were an upper bound of $\{a \wedge y_k : a \in A\}$. Then $z_i + w - y_k < z_i$ would be an upper bound of A below the supremum. Indeed, if $a \in A$, then

$$a = a - a \wedge y_k + a \wedge y_k \leqslant a - a \wedge y_k + w$$

and

$$a - a \wedge y_k = |a \wedge z - a \wedge y_k| \leqslant |z - y_k| = z - y_k.$$

□

We now analyze the number of iterations of order adherence that might be needed to attain the order closure.

Theorem 2.6. [∀solid_sets_require_arbitrarily_many_iterations] *For every cardinal κ with $\aleph_0 \leq \kappa$ and every ordinal $\xi \leq \kappa^+$ there exists a solid set S generated by a set of size κ such that ξ many iterations of order adherence are needed to reach its order closure.*

Proof. As a matter of notation, throughout this proof superscripts will be used to denote Cartesian products of sets. For example, I^ω is the set of all tuples $(i_n)_{n<\omega}$ with $i_n \in I$ for all n. But when the base is ω, it will mean ordinal power, not Cartesian product. For example, $\omega^2 = \sup\{\omega, \omega + \omega, \ldots\}$ is a countable ordinal, not a set of tuples.

We define a partial order relation on ordinals. Suppose that we are given ordinals $\zeta \leq \zeta'$. Let us write them in Cantor normal form

$$\zeta = \sum_{i=1}^{n} \omega^{\beta_i} c_i \qquad \text{and} \qquad \zeta' = \sum_{i=1}^{m} \omega^{\gamma_i} d_i.$$

We say that $\zeta \prec \zeta'$ if $m \leq n$, $\beta_i = \gamma_i$, $c_i = d_i$ for $i < m$, and either $\gamma_m = \beta_m$ and $c_m < d_m$, or $\beta_m < \gamma_m < \beta_{m-1}$; if $m = 1$ there is no β_{m-1}, and the last condition must be interpreted as $\beta_m < \gamma_m$. Some basic properties are:

P1. $\preceq$ is a partial order contained in the usual ordinal order $\leq$.

P2. For every ζ, the relations $\prec$ and $<$ coincide on $\{\zeta' : \zeta \prec \zeta'\}$.

Now fix an ordinal ξ. Consider the compact space

$$K = \{(x_\zeta)_{\zeta \leq \omega^\xi + 1} : \zeta \prec \zeta' \Rightarrow x_\zeta \leq x_{\zeta'}\} \subset \{0,1\}^{\omega^\xi + 1}.$$

Let $\pi_\zeta \in C(K)$ be the projection onto the ζ-th coordinate, and let S be the solid set of $C(K)$ generated by the π_ζ for ζ a successor ordinal. We claim that ξ many iterations of the order adherence are needed to reach the order closure of S. This proves the statement for $\xi < \kappa^+$. We will deal with the special case $\xi = \kappa^+$ at the end of the proof.

By the definition of K, we have $\pi_\zeta \leq \pi_{\zeta'}$ whenever $\zeta \prec \zeta'$. Moreover:

Claim 1: If Z is an infinite pairwise $\prec$-incomparable family of ordinals, then $\inf\{\pi_\zeta : \zeta \in Z\} = 0$.

Proof of claim: Suppose $0 \leq f$ is a lower bound. In particular this means that $f(x) = 0$ whenever $x_\zeta = 0$ for some $\zeta \in Z$. Thus, it is enough to prove that

$$D = \{x \in K : \exists \zeta \in Z\ x_\zeta = 0\}$$

is dense in K. So take $y \in K$ and a basic neighborhood of the form

$$W = \{x \in K : x|_F = y|_F\}$$

for finite $F \subset \omega^\xi + 1$. We must find $x \in W \cap D$. There must exist $\zeta_0 \in Z$ such that $\zeta' \not\preceq \zeta_0$ for all $\zeta' \in F$ because elements $\prec$-above ζ' are linearly ordered by $\prec$. Define x by declaring $x_\zeta = 0$ if $\zeta \preceq \zeta_0$ and $x_\zeta = y_\zeta$ otherwise.

Claim 2: If Z is a nonempty $\prec$-totally ordered family of ordinals with $\sup(Z) = \alpha$, then

$$\sup\{\pi_\zeta : \zeta \in Z\} = \pi_\alpha.$$

Proof of claim: It is clear that π_α is an upper bound. The nontrivial case is when the supremum is not a maximum. Let $f \in C(K)$ be another upper bound. If $\pi_\alpha \not\leq f$, then there exists $x \in K$ such that $f(x) < x_\alpha$. Since $f \geq 0$, we must have $x_\alpha = 1$. By continuity in the product topology, there exists a finite set $F \subseteq \omega^\xi$ such that $y|_F = x|_F$ implies $f(y) < 1$. Take $\zeta_0 \in Z$ such that $\max(F \cap \alpha) < \zeta_0$. Consider $y \in K$ such that $y_\zeta = 1$ if $\zeta_0 \preceq \zeta$ and $y_\zeta = x_\zeta$ otherwise. Then

$$1 = y_{\zeta_0} \leq f(y) < 1,$$

a contradiction.

Let Z_β be the set of all ordinals $\zeta \leq \omega^\xi$ whose minimal exponent in Cantor normal form is strictly less than β in the usual ordinal order. Thus Z_1 is the set of all successor ordinals. Let

$$S_\beta = \{f \in C(K) : |f| \leq \pi_\zeta \text{ for some } \zeta \in Z_\beta\}.$$

Claim 3: $o\text{-}adh^\beta(S) = S_{1+\beta}$.

Proof of claim: We prove it by induction on β. For $\beta = 0$, this is the definition of S. The limit case is trivial. For the successor case, it is enough to prove that

$$o\text{-}adh(S_\gamma) = S_{\gamma+1}$$

for every $\gamma \leq \omega^\xi + 1$. The inclusion $[\supseteq]$ follows from Claim 2 because every $\zeta \in Z_{\gamma+1}$ can be found as a supremum of a $\prec$-increasing set from Z_γ. For the inclusion $[\subseteq]$, suppose that we have an upward directed net $\{g_i\} \subseteq S_\gamma$ such that $|f| = \sup_i g_i$. For every i we know that

$$A_i := \{\zeta \in Z_\gamma : g_i \leq \pi_\zeta\} \neq \emptyset.$$

Let M_i be the set of $\prec$-minimal elements of A_i. Since it extends the ordinal order, $\prec$ is a well-founded relation, which implies that every element of A_i is $\prec$-above an element of M_i. Moreover, by Claim 1, each M_i is finite. If $i \leq j$ then $A_j \subseteq A_i$, and hence, since $\prec$ is a linear order above any given element, $|M_j| \leq |M_i|$. Therefore, by restricting from an index onward, we may suppose that all sets M_i have the same cardinality k and write $M_i = \{\mu_i^1, \ldots, \mu_i^k\}$ in such a way that $\mu_i^t \preceq \mu_j^t$ when $i \leq j$. But then, since $g_i \leq \pi_{\mu_i^1}$, we have

$$|f| \leq \sup_i \pi_{\mu_i^1} = \pi_{\sup_i \mu_i^1}$$

and $\sup_i \mu_i^1 \in Z_{\gamma+1}$.

After Claim 3, it is enough to notice that $\pi_{\omega^\gamma} \in S_{\gamma+1} \setminus S_\gamma$. It is trivial that $\pi_{\omega^\gamma} \in S_{\gamma+1}$. If we had $\pi_{\omega^\gamma} \in S_\gamma$, then $\pi_{\omega^\gamma} \leq \pi_\zeta$ for some $\zeta \in Z_\gamma$. We would have $\omega^\gamma \not\preceq \zeta$. If we consider $x \in K$ with $x_\alpha = 0$ if $\alpha \preceq \zeta$ and $x_\alpha = 1$ otherwise, then

$$\pi_{\omega^\gamma}(x) = 1 < 0 = \pi_\zeta(x),$$

a contradiction. This finishes the case $\xi < \kappa^+$.

For the case of κ^+, consider, for every $\xi < \kappa^+$, the solid set S_ξ generated by a set $\{f_\xi^\alpha : \alpha < \kappa\}$ of elements of norm one in the Banach lattice X_ξ constructed above, where ξ many iterations of order adherence were needed to reach the order closure. Take X to be the ℓ_∞-sum of all those X_ξ, and let S be the solid set generated by the elements $f_\alpha = (f_\alpha^\xi)_{\xi<\kappa^+}$. □

The ordinal κ^+ in Theorem 2.6 is likely not optimal. Although we will not pursue this issue in detail, we include a lemma that might be useful to tackle this question.

Lemma 2.7. [∀ solid_generated_orderAdherence] *If $S = \operatorname{solid}(G)$ for some set G of positive elements, then*

$$S^o = \left\{ x : \exists \text{ net } (a_i) \subset G \text{ with } |x| = o\text{-}\lim_{i \in I} |x| \wedge a_i = \sup_{i \in I} |x| \wedge a_i \right\}.$$

Proof. A first observation is that, for a given x, the fact that

$$|x| = o\text{-}\lim_{i \in I} |x| \wedge a_i = \sup_{i \in I} |x| \wedge a_i$$

is equivalent to the simultaneous satisfaction of

$$x^+ = o\text{-}\lim_{i \in I} x^+ \wedge a_i = \sup_{i \in I} x^+ \wedge a_i,$$

$$x^- = o\text{-}\lim_{i \in I} x^- \wedge a_i = \sup_{i \in I} x^- \wedge a_i.$$

For ($\Leftarrow$) use that $|x| = x^+ + x^-$, and for ($\Rightarrow$) use the property checked at the end of Proposition 2.5.

Now we proceed to prove the equality of sets. The inclusion [$\supseteq$] follows from the previous observation. For [$\subseteq$], suppose $x = o\text{-}\lim_{i \in I} x_i$ with $x_i \in S$. Then $|x| = o\text{-}\lim |x_i|$. In particular, there is an increasing net $\{y_j\}$, which we may suppose consists of positive elements, which is eventually below $|x_i|$ and has supremum $|x|$. For every j, we may thus choose a_j such that $y_j \leq a_j \in G$. ☐

3. A simpler solution to Fremlin's problem on weakly Fatou norms

Recall that a lattice norm $\|\cdot\|$ on a vector lattice X has the *Fatou property* if

$$\|x\| = \sup_{\alpha} \|x_\alpha\|$$

whenever $0 \leqslant x_\alpha \uparrow x$ in X, and it is *weakly Fatou* if there exists $K \geqslant 1$ such that

$$\|x\| \leqslant K \sup_{\alpha} \|x_\alpha\|$$

whenever $0 \leqslant x_\alpha \uparrow x$ in X.

The goal of this section is to prove the following theorem.

Theorem 3.1. [∀ exists_weaklyFatou_not_equivalent_fatou] *There exists a Banach lattice with a weakly Fatou norm which is not equivalent to any lattice norm with the Fatou property.*

The proof has two parts. For each $n \in \mathbb{N}$ we construct a separable Banach lattice X_n such that

(i) X_n is weakly Fatou with constant 2;
(ii) the n-th iterated order adherence of the unit ball of X_n contains an element of norm 2^n.

We then take the c_0-sum of the spaces X_n. If that sum admitted an equivalent Fatou norm, then every iterated order adherence of its unit ball would stay uniformly bounded, contradicting (ii).

We will try to make the construction as explicit as possible. We will consider the finite-height tree

$$G_n := \bigcup_{k=0}^{n} \mathbb{N}^k,$$

and characteristic functions of simple cylinder sets in the product space $\mathbb{N}^{H_n}$, where $H_n = G_n \backslash \mathbb{N}^n$ is the set of non-terminal nodes in G_n. This provides a family of functions in which every parent will be the supremum of its children, while a certain norm will still assign the value 2^{-k} to every node on level k. Establishing the uniform weak Fatou estimate for this norm will be the most technical part of the argument.

Let us begin with some simple properties of order adherence.

Lemma 3.2. [$\forall$ `iteratedOrderAdherence_mono_and_scale`] *Let $A, B \subseteq X$ be solid and let $\lambda > 0$.*

(a) *If $A \subseteq B$, then $A^{o(m)} \subseteq B^{o(m)}$ for all $m \geqslant 0$.*
(b) *One has $(\lambda A)^o = \lambda A^o$, hence $(\lambda A)^{o(m)} = \lambda A^{o(m)}$ for all $m \geqslant 0$.*

Proof. These claims are essentially immediate from the definition. □

The following lemma translates the Fatou and weak Fatou properties into statements about how much the unit ball can expand after iterated order adherence.

Lemma 3.3. [$\forall$ `weakFatou_iterated_unitBall` and $\forall$ `fatou_iterated_unitBall`] *Let $(X, \|\cdot\|)$ be a normed vector lattice.*

(a) *If $\|\cdot\|$ is weakly Fatou with constant K, then*

$$B_X^{o(m)} \subseteq K^m B_X \qquad (m \geqslant 1).$$

(b) *If $\|\cdot\|$ has the Fatou property, then*

$$B_X^{o(m)} = B_X \qquad (m \geqslant 1).$$

Proof. It is enough to treat one order adherence step.

For (a), let $x \in B_X^o$. Choose an upward directed $B \subseteq B_X \cap X_+$ with $|x| \leqslant \sup B$. By the weak Fatou property, we have

$$\|x\| \leqslant \|\sup B\| \leqslant K \sup_{b \in B} \|b\| \leqslant K.$$

Hence, $B_X^o \subseteq K B_X$ and iteration gives the claim.

For (b), the inclusion $B_X \subseteq B_X^o$ is trivial. Conversely, if $x \in B_X^o$, choose B as above. The Fatou property gives

$$\|\sup B\| = \sup_{b \in B} \|b\| \leqslant 1,$$

so again $\|x\| \leqslant 1$. Thus, $B_X^o = B_X$ and the iterated identity follows. □

In our construction, it will be enough to work with a sequential version of Fatou/weakly Fatou norms. For convenience, we recall the following definitions.

Definition 3.4. Let $(X, \|\cdot\|)$ be a normed vector lattice and let $K \geqslant 1$.

(a) We say that K is a *weak sequential Nakano constant* for X if for every increasing sequence (x_m) in X_+ with an upper bound in X and with $\sup_m \|x_m\| \leqslant 1$, and for every $\varepsilon > 0$, there exists an upper bound $y \in X_+$ of the sequence such that $\|y\| \leqslant K + \varepsilon$.
(b) We say that K is a *weak Nakano constant* for X if the same statement holds with increasing sequences replaced by upward directed subsets of the unit ball of X_+ having an upper bound in X.

We shall use the following elementary reduction essentially based in [2, Lemma 2.5], which we include for completeness.

Proposition 3.5. [∀ `separable_weakSequentialNakano_implies_weakNakano`] *Let Y be a separable normed vector lattice and $K \geqslant 1$ a weak sequential Nakano constant for Y. Then K is a weak Nakano constant for Y. In particular, Y is weakly Fatou with constant K.*

Proof. Let $A \subseteq Y_+$ be upward directed, order bounded, and satisfy $\sup_{a \in A}\|a\| \leqslant 1$. Since Y is separable, A has a countable dense subset D. Let C be the set of all finite suprema of elements of D. Then C is countable, upward directed, and has the same upper bounds as A: every upper bound of A clearly bounds C, while if y bounds D then $y - d \in Y_+$ for every $d \in D$, hence also $y - a \in Y_+$ for every $a \in A$ by norm closedness of the positive cone. Enumerating C as (c_m) and replacing it by the increasing sequence $d_m := c_1 \vee \cdots \vee c_m$, we obtain an increasing sequence with the same set of upper bounds as A. By hypothesis, for every $\varepsilon > 0$ there is $y \in S$ such that $d_m \leqslant y$ for all m and $\|y\| \leqslant K + \varepsilon$. Hence y is an upper bound of A. This is exactly K being a weak Nakano constant for Y. □

Let us now introduce the notation for the underlying tree and corresponding space of functions on it. We fix $n \in \mathbb{N}$ and put

$$G_n := \bigcup_{k=0}^{n} \mathbb{N}^k, \qquad H_n := \bigcup_{k=0}^{n-1} \mathbb{N}^k.$$

We will refer to H_n as the set of non-terminal nodes of the tree G_n. The unique element of $\mathbb{N}^0$ is denoted by $\varnothing$ and will be called the root. If $t \in \mathbb{N}^k$ and $m \in \mathbb{N}$, write $t^\frown m \in \mathbb{N}^{k+1}$ for concatenation. We define the parent map $\pi : G_n \to G_n$ by

$$\pi(\varnothing) := \varnothing, \qquad \pi(t^\frown m) := t.$$

For $t = (t_1, \ldots, t_k)$ and $0 \leqslant j \leqslant k$, write $t|j := (t_1, \ldots, t_j)$, with $t|0 = \varnothing$. Let

$$I_n := \mathbb{N}^{H_n},$$

equipped with the product topology, each copy of $\mathbb{N}$ carrying the discrete topology. An element $\alpha \in I_n$ can be considered as a function which assigns to each non-terminal node $u \in H_n$ a natural number $\alpha(u)$.

For $t \in G_n$ of length k, define the *cylinder*

$$E_t := \{\alpha \in I_n : \alpha(t|j-1) \leqslant t_j \text{ for } 1 \leqslant j \leqslant k\},$$

and set $E_\varnothing := I_n$. Let $C_b(I_n)$ denote the space of bounded continuous functions on I_n and for $t \in G_n$ let us consider the corresponding characteristic function

$$s_t := \chi_{E_t}.$$

Lemma 3.6. [∀ `treeCylinder_isClopen` and ∀ `treeFunction_child_properties`] *For every $t \in G_n$, the following properties hold.*

(a) *E_t is clopen in I_n, hence $s_t \in C_b(I_n)$;*
(b) *if $|t| < n$ and $m \in \mathbb{N}$, then $E_{t^\frown m} \subseteq E_t$, hence $s_{t^\frown m} \leqslant s_t$;*
(c) *if $|t| < n$, then*

$$s_t = \sup_{m \in \mathbb{N}} s_{t^\frown m}$$

pointwise on I_n.

Proof. Part (a) is immediate because E_t is defined by finitely many coordinate inequalities.

Part (b) is immediate from the definition.

For (c), note that

$$E_{t^\frown m} = E_t \cap \{\alpha \in I_n : \alpha(t) \leqslant m\}.$$

As m increases these sets are increasing, and their union is E_t. Taking characteristic functions gives the claim. $\square$

The following combinatorial fact will play a key role in the construction.

Lemma 3.7. [$\forall$`treeCylinder_finite_cover`] *Let $t \in G_n$ and let $F \subseteq G_n$ be finite. If*

$$E_t \subseteq \bigcup_{u\in F} E_u,$$

then there exists $u \in F$ with $E_t \subseteq E_u$.

Proof. Assume that $E_t \not\subseteq E_u$ for every $u \in F$. We shall construct $\alpha \in E_t \setminus \bigcup_{u\in F} E_u$.

For every proper ancestor $t|j$ of t with $0 \leqslant j < |t|$, set

$$\alpha(t|j) := t_{j+1}.$$

Now let $a \in H_n$ be any non-terminal node which is not a proper ancestor of t but is a parent of some element of F. Since F is finite, the set

$$M_a := \{m \in \mathbb{N} : a^\frown m \text{ is an initial segment of some } u \in F\}$$

is finite. Choose $\alpha(a) > \max M_a$. On all remaining coordinates set $\alpha(a) := 1$. By construction, $\alpha \in E_t$. Take $u = (u_1, \dots, u_k) \in F$. Since $E_t \not\subseteq E_u$, one of the following two cases occurs.

If the parent $u|k-1$ is a proper ancestor of t, then necessarily $k \leqslant |t|$ and $u_i = t_i$ for $1 \leqslant i \leqslant k-1$. If $u_k \geqslant t_k$, then every point of E_t satisfies the defining inequalities for E_u, contrary to the assumption that $E_t \not\subseteq E_u$. Hence $u_k < t_k$, and therefore

$$\alpha(u|k-1) = t_k > u_k.$$

So, $\alpha \notin E_u$.

If the parent $u|k-1$ is not a proper ancestor of t, then by construction $u|k-1$ belongs to the exceptional set of coordinates on which α was chosen to dominate all child labels appearing in F. In particular,

$$\alpha(u|k-1) > u_k,$$

so again $\alpha \notin E_u$.

Thus $\alpha \notin E_u$ for every $u \in F$, a contradiction. $\square$

The next lemma will later allow us to ignore pieces supported on bands that do not recur. To make this precise, we need a definition.

Definition 3.8. Two subsets $A, B \subseteq G_n$ are *π-disjoint* if $\pi[A] \cap \pi[B] = \varnothing$.

Lemma 3.9. [$\forall$`parentDisjoint_treeFunctions_iInf`] *Let (F_m) be a sequence of finite subsets of G_n which are pairwise π-disjoint. Then*

$$\inf_{m\in\mathbb{N}} \sup_{t\in F_m} s_t = 0$$

in the vector lattice $C_b(I_n)$.

Proof. Put $g_m := \sup_{t\in F_m} s_t = \chi_{U_m}$, where $U_m := \bigcup_{t\in F_m} E_t$. It suffices to show that $\bigcap_m U_m$ has empty interior.

Suppose, towards a contradiction, that some non-empty basic open cylinder $\Omega \subseteq I_n$ is contained in every U_m. Then Ω fixes only finitely many coordinates, say those in a finite set $Q \subseteq H_n$.

Because the sets $\pi[F_m]$ are pairwise disjoint and finite, we can choose m such that $\pi[F_m] \cap Q = \varnothing$ and $\varnothing \notin F_m$.

Choose any $\alpha \in \Omega$. For every $u \in \pi[F_m]$, change the value of $\alpha(u)$ to a number strictly larger than every child label appearing above u in F_m. Since no such u lies in Q, the modified point still belongs to Ω; denote it by β.

Now take $t \in F_m$ and let $u := \pi(t)$. By construction, $\beta(u) > t_{|t|}$, so $\beta \notin E_t$. Hence $\beta \notin U_m$, which contradicts $\Omega \subseteq U_m$. □

We are now in position to define a lattice norm with the required properties. Let $W_n := c_{00}(G_n)$, the lattice of finitely supported real functions on G_n. For $t \in G_n$, write e_t for the characteristic function of the singleton $\{t\}$.

For $w \in W_n$, we define the weight

$$\rho_n(w) := \sum_{t\in G_n} 2^{-|t|}\,|w(t)|.$$

Also, let us consider the linear map $T_n : W_n \to C_b(I_n)$ given by

$$T_n w := \sum_{t\in G_n} w(t) s_t$$

for $w \in W_n$. For $x \in C_b(I_n)$, set

$$p_n(x) := \inf\{\rho_n(w) : w \in W_n^+,\ |x| \leqslant T_n w\}.$$

Lemma 3.10. [∀`treeLatticeSeminorm`] *p_n is a lattice seminorm on $C_b(I_n)$.*

Proof. Monotonicity and the lattice property are immediate from the definition. Positive homogeneity is obvious.

For subadditivity, take $x, y \in X_n$ and $\varepsilon > 0$. Choose $u, v \in W_n^+$ with $|x| \leqslant T_n u$, $|y| \leqslant T_n v$, $\rho_n(u) < p_n(x) + \varepsilon$, and $\rho_n(v) < p_n(y) + \varepsilon$. Then

$$|x + y| \leqslant |x| + |y| \leqslant T_n(u + v),$$

so

$$p_n(x + y) \leqslant \rho_n(u + v) = \rho_n(u) + \rho_n(v) < p_n(x) + p_n(y) + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ completes the proof. □

Lemma 3.11. [∀`treeSeminorm_norm_comparison`] *For every $x \in C_b(I_n)$ one has*

$$p_n(x) \leqslant \|x\|_\infty \leqslant 2^n p_n(x).$$

Proof. Because $|x| \leqslant \|x\|_\infty s_\varnothing$ and $\rho_n(e_\varnothing) = 1$, the first inequality is immediate.

For the second, let $w \in W_n^+$ satisfy $|x| \leqslant T_n w$. Since each s_t is $\{0, 1\}$-valued, we have

$$\|x\|_\infty \leqslant \|T_n w\|_\infty \leqslant \sum_{t\in G_n} w(t) \leqslant 2^n \sum_{t\in G_n} 2^{-|t|} w(t) = 2^n \rho_n(w).$$

Taking the infimum over all such w proves the claim. □

The next lemma gives the exact values of p_n on the tree functions.

Lemma 3.12. [∀ treeSeminorm_exact_basis] *Let $t \in G_n$. If $w \in W_n^+$ satisfies $s_t \leqslant T_n w$, then $\rho_n(w) \geqslant 2^{-|t|}$. Consequently,*

$$p_n(s_t) = 2^{-|t|}.$$

Proof. The upper bound $p_n(s_t) \leqslant 2^{-|t|}$ is given by $w = e_t$.

For the reverse inequality, let $w \in W_n^+$ satisfy $s_t \leqslant T_n w$, and let

$$U := \{u \in \operatorname{supp} w : E_t \subseteq E_u,\ u \neq t\}.$$

Move all coefficients supported on U down to t:

$$w' := w + \Big(\sum_{u \in U} w(u)\Big) e_t - \sum_{u \in U} w(u) e_u.$$

If $\alpha \in E_t$, then $E_t \subseteq E_u$ for every $u \in U$, so $T_n w'(\alpha) = T_n w(\alpha) \geqslant 1$. Thus $s_t \leqslant T_n w'$. Moreover, every $u \in U$ satisfies $|u| \leqslant |t|$. Indeed, if $|u| > |t|$, then the last defining inequality for E_u involves a coordinate which is not constrained by membership in E_t, so $E_t \subseteq E_u$ is impossible. Hence $2^{-|t|} \leqslant 2^{-|u|}$ and $\rho_n(w') \leqslant \rho_n(w)$.

No strict upper bound of t remains in $\operatorname{supp} w'$. Therefore, Lemma 3.7, applied to the finite family $\operatorname{supp} w' \setminus \{t\}$, shows that

$$E_t \not\subseteq \bigcup_{u \in \operatorname{supp} w',\, u \neq t} E_u.$$

Choose $\alpha \in E_t$ outside that union. Evaluating at α gives

$$1 = s_t(\alpha) \leqslant T_n w'(\alpha) = w'(t),$$

because every term other than t vanishes at α. Hence

$$\rho_n(w) \geqslant \rho_n(w') \geqslant 2^{-|t|} w'(t) \geqslant 2^{-|t|}.$$

This proves the claim. □

Let X_n be the closed sublattice of $C_b(I_n)$ generated by the countable family $\{s_t : t \in G_n\}$, equipped with the restriction of p_n.

Corollary 3.13. [∀ component_basic] *(X_n, p_n) is a separable Banach lattice. Moreover, $p_n(\mathbf{1}) = 1$ and every terminal node $t \in \mathbb{N}^n$ satisfies $p_n(s_t) = 2^{-n}$.*

Proof. Separability is clear from the countable generating family, and completeness follows from Lemma 3.11 because X_n is closed in the supremum norm. The formulae for $p_n(\mathbf{1})$ and the terminal nodes are immediate from Lemma 3.12. □

We now focus on the technical part of estimating the weak Fatou constant of (X_n, p_n).

For each non-terminal node $u \in H_n$, let

$$C_u := \{u^\frown m : m \in \mathbb{N}\},$$

and let $B_u \subseteq W_n$ be the band of functions supported on C_u. Also put $B_\varnothing := \mathbb{R} e_\varnothing$. Then

$$W_n = B_\varnothing \oplus \bigoplus_{u \in H_n} B_u$$

as an algebraic ℓ^1-sum with respect to ρ_n.

For each such band B, let $P_B : W_n \to B$ denote the coordinate projection. If Λ is a finite family of these bands, write

$$P_\Lambda := \sum_{B\in\Lambda} P_B.$$

Define the upshift $S_n : W_n \to W_n$ by

$$S_n e_\varnothing := e_\varnothing, \qquad S_n e_{t^\frown m} := e_t.$$

Lemma 3.14. [∀`treeUpshift_basic`] *For every $w \in W_n^+$ one has*

$$\rho_n(S_n w) \leqslant 2\rho_n(w) \qquad \textit{and} \qquad T_n w \leqslant T_n S_n w.$$

Proof. The first inequality follows from $2^{-|\pi(t)|} = 2 \cdot 2^{-|t|}$ for $t \neq \varnothing$.

For the second, it is enough to check it on the basis vectors $e_{t^\frown m}$. There it reads $s_{t^\frown m} \leqslant s_t$, which is Lemma 3.6(b). The claim follows by linearity and positivity. ☐

Let (x_m) be an increasing sequence in X_n^+ with $p_n(x_m) \leqslant 1$ for all m. Fix $\varepsilon > 0$.

For each m choose $w_m \in W_n^+$ such that

$$x_m \leqslant T_n w_m \qquad \text{and} \qquad \rho_n(w_m) < 1 + \varepsilon/4.$$

Since (x_m) is increasing, every subsequence of $(T_n w_m)$ still dominates the whole sequence (x_m).

Lemma 3.15. [∀`tree_sharp_subsequence`] *After passing to a subsequence, we may suppose that for every band $B \in \{B_\varnothing\} \cup \{B_u : u \in H_n\}$ the sequence $\big(\rho_n(P_B w_m)\big)$ converges.*

Proof. There are only countably many bands, and each projected norm is bounded by $\rho_n(w_m)$. A standard diagonal argument gives the claim. ☐

Fix such a subsequence and write

$$\lambda_B := \lim_{m\to\infty} \rho_n(P_B w_m)$$

for each band B. Since ρ_n is the ℓ^1-sum of the band norms, we have

$$\sum_B \lambda_B \leqslant \liminf_{m\to\infty} \rho_n(w_m) \leqslant 1 + \varepsilon/4.$$

Lemma 3.16. [∀`tree_trim`] *There exists a new sequence $(\widetilde{w}_m)$ in W_n^+ such that:*

(i) *$x_m \leqslant T_n \widetilde{w}_m$ for all m;*
(ii) *$\rho_n(\widetilde{w}_m) < 1 + \varepsilon/2$ for all m;*
(iii) *only finitely many bands in the decomposition given above meet $\operatorname{supp} \widetilde{w}_m$ for infinitely many values of m.*

Proof. Put $M := 2^n$. Choose a finite family Λ of bands, containing $B_\varnothing$, such that

$$\sum_{B\notin\Lambda} \lambda_B < \frac{\varepsilon}{8M}.$$

Enumerate the remaining bands as $(D_j)_{j\in\mathbb{N}}$ and set

$$\mu_j := \lambda_{D_j} + \frac{\varepsilon}{16M\, 2^j}.$$

Then

$$\sum_{j=1}^{\infty} \mu_j < \frac{\varepsilon}{4M}.$$

For each j choose $N_j \in \mathbb{N}$ such that

$$\rho_n(P_{D_j} w_m) < \mu_j \qquad (m \geqslant N_j).$$

After increasing the N_j if necessary, we may assume that (N_j) is strictly increasing.

Now define

$$\widetilde{w}_m := w_m - \sum_{j\colon N_j \leqslant m} P_{D_j} w_m + M\Big(\sum_{j\colon N_j \leqslant m} \mu_j\Big) e_\varnothing.$$

This is a positive vector. Let $z \in B^+_{D_j}$ with $\rho_n(z) \leqslant \mu_j$. Since every coefficient of z has weight at least $2^{-n} = M^{-1}$, we have

$$\sum_{t \in G_n} |z(t)| \leq 2^n \sum_{t \in G_n} 2^{-|t|} |z(t)| \leq M\mu_j.$$

Therefore

$$T_n z \leqslant M\mu_j\, \mathbf{1} = T_n(M\mu_j e_\varnothing).$$

Applying this with $z = P_{D_j} w_m$ shows that $T_n \widetilde{w}_m \geqslant T_n w_m \geqslant x_m$. Also,

$$\rho_n(\widetilde{w}_m) \leqslant \rho_n(w_m) + M \sum_{j=1}^{\infty} \mu_j < 1 + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = 1 + \frac{\varepsilon}{2}.$$

Finally, if $D_j \notin \Lambda$ and $m \geqslant N_j$, then $P_{D_j} \widetilde{w}_m = 0$. Hence, only bands in Λ can occur infinitely often. $\square$

Replace (w_m) by the sequence given by Lemma 3.16. Let Λ be the finite family of bands that still occur infinitely often.

Lemma 3.17. [$\forall$ `tree_thinning`] *After passing to a further subsequence, we may suppose that every band $B \notin \Lambda$ meets the support of w_m for at most one value of m.*

Proof. For every band $B \notin \Lambda$, let $\ell(B)$ be the last index m for which $P_B w_m \neq 0$.

Choose $m_1 < m_2 < \cdots$ inductively as follows. Having chosen m_k, let $\mathcal{F}_k$ be the finite set of bands $B \notin \Lambda$ with $P_B w_{m_k} \neq 0$, and choose

$$m_{k+1} > \max\big(\{m_k\} \cup \{\ell(B) : B \in \mathcal{F}_k\}\big).$$

Then every band outside Λ which appears at stage m_k never appears again. Passing to the subsequence (w_{m_k}) proves the claim. $\square$

Write

$$u_m := P_\Lambda w_m, \qquad v_m := w_m - u_m.$$

By construction, the bands meeting $\operatorname{supp} v_m$ are disjoint from those meeting $\operatorname{supp} v_{m'}$ whenever $m \neq m'$.

Lemma 3.18. [$\forall$ `tree_transient`] *The finite sets $F_m := \operatorname{supp} v_m$ are pairwise π-disjoint. Consequently, the sequence $(T_n v_m)$ has no non-zero common lower bound.*

Proof. If $s, t \in G_n \setminus \{\varnothing\}$ belong to the same sibling class, then $\pi(s) = \pi(t)$. Thus, two supports can share a parent only if they meet the same band B_u for some $u \in H_n$. By Lemma 3.17, this never happens for distinct m, so the sets F_m are pairwise π-disjoint.

Let $z \in X_n^+$ be a lower bound of all $T_n v_m$. Put $C := 2^n \sup_m \rho_n(v_m)$. Since every coefficient has weight at least 2^{-n}, we have

$$T_n v_m \leqslant C \sup_{t \in F_m} s_t.$$

If $C = 0$, then every v_m is zero and there is nothing to prove. Hence,

$$C^{-1} z \leqslant \sup_{t \in F_m} s_t \qquad (m \in \mathbb{N}).$$

Lemma 3.9 now gives $z = 0$. ☐

The preceding lemmas allow us to finally bound the weak Fatou constant of (X_n, p_n).

Proposition 3.19. [∀ `component_moderated`] *For every increasing sequence (x_m) in X_n^+ with $p_n(x_m) \leqslant 1$ and for every $\varepsilon > 0$, there exists $y \in T_n[W_n^+]$ such that*

$$x_m \leqslant y \quad (m \in \mathbb{N}), \qquad p_n(y) \leqslant 2 + \varepsilon.$$

Proof. Start with the sequence constructed above and apply Lemmas 3.15 to 3.17. Whenever we pass to a subsequence of (w_m), we pass to the corresponding subsequence of (x_m) and relabel. Since the original sequence (x_m) is increasing, any upper bound for the relabelled subsequence is also an upper bound for the original sequence.

The supports of $S_n u_m$ lie in the finite set consisting of the root together with the parents of the finitely many bands in Λ. Hence $(S_n u_m)$ lives in a finite-dimensional subspace of W_n. By boundedness and Lemma 3.14, after passing to another subsequence we may assume that $S_n u_m$ converges in W_n to some $u \in W_n^+$. Put $y := T_n u$. Again by Lemma 3.14, we have

$$\rho_n(u) = \lim_m \rho_n(S_n u_m) \leqslant 2 \sup_m \rho_n(u_m) \leqslant 2 + \varepsilon,$$

so

$$p_n(y) \leqslant \rho_n(u) \leqslant 2 + \varepsilon.$$

Since T_n is continuous on the finite-dimensional space containing all $S_n u_m$, we have

$$\|T_n S_n u_m - y\|_\infty \longrightarrow 0.$$

Fix $j \in \mathbb{N}$ and $\eta > 0$. Choose m_0 such that

$$\|T_n S_n u_m - y\|_\infty \leqslant \eta \qquad (m \geqslant m_0).$$

For $m \geqslant \max\{j, m_0\}$ we have

$$x_j \leqslant x_m \leqslant T_n w_m = T_n u_m + T_n v_m \leqslant T_n S_n u_m + T_n v_m \leqslant y + \eta \mathbf{1} + T_n v_m.$$

Hence

$$(x_j - y - \eta \mathbf{1})^+ \leqslant T_n v_m \qquad (m \geqslant \max\{j, m_0\}).$$

By Lemma 3.18, this tail has no non-zero common lower bound. Therefore $(x_j - y - \eta\mathbf{1})^+ = 0$, that is, $x_j \leqslant y + \eta\mathbf{1}$. Letting $\eta \downarrow 0$ gives $x_j \leqslant y$. Since j was arbitrary, y dominates the whole sequence. ☐

Corollary 3.20. [∀ `component_weakFatou`] *The Banach lattice (X_n, p_n) has weak Nakano constant 2, hence weak Fatou constant 2.*

Proof. The positive set $T_n[W_n^+]$ satisfies the hypothesis of Proposition 3.5 by Proposition 3.19. Since X_n is separable, the conclusion follows. ☐

Let now

$$L_n := \{s_t : t \in \mathbb{N}^n\}$$

be the set of characteristic functions of terminal nodes of the tree G_n.

Proposition 3.21. [∀ `component_large_iterated_adherence`] *For each $n \in \mathbb{N}$:*

(i) *(X_n, p_n) is a separable Banach lattice with weak Fatou constant 2;*
(ii) *the constant function $\mathbf{1}$ satisfies $2^n\mathbf{1} \in B_{X_n}^{o(n)}$;*
(iii) *$p_n(2^n\mathbf{1}) = 2^n$.*

Proof. Part (i) is Corollaries 3.13 and 3.20.

For (ii), Lemma 3.12 gives $p_n(2^n s_t) = 1$ for every terminal t, so the solid hull

$$A_n := \operatorname{sol}(2^n L_n)$$

is contained in B_{X_n}. We claim, more generally, that for every node $t \in G_n$,

$$2^n s_t \in A_n^{o(n-|t|)}.$$

If $|t| = n$, this is tautological. If $|t| < n$, then by Lemma 3.6(c),

$$2^n s_t = \sup_{m\in\mathbb{N}} 2^n s_{t^\frown m}.$$

The children form an increasing sequence, so $2^n s_t \in \left(A_n^{o(n-|t|-1)}\right)^o = A_n^{o(n-|t|)}$. Taking $t = \varnothing$ yields $2^n\mathbf{1} \in A_n^{o(n)} \subseteq B_{X_n}^{o(n)}$.

Part (iii) follows from $p_n(\mathbf{1}) = 1$. ☐

Finally, let us consider the space

$$X := c_0(X_n)_{n\in\mathbb{N}},$$

equipped with the usual supremum norm

$$\|(x_n)\| := \sup_n p_n(x_n).$$

Lemma 3.22. [∀ `finalSpace_weakFatou`] *X is a Banach lattice with weak Fatou constant 2.*

Proof. Completeness is standard for c_0-sums. Let $0 \leqslant x_\alpha \uparrow x$ in X. For each n, the nth coordinate satisfies

$$p_n(x(n)) \leqslant 2\sup_\alpha p_n(x_\alpha(n)) \leqslant 2\sup_\alpha \|x_\alpha\|.$$

Taking the supremum over n gives

$$\|x\| \leqslant 2\sup_\alpha \|x_\alpha\|.$$

☐

For each n, let $z_n \in X$ be the vector whose nth coordinate is $2^n\mathbf{1} \in X_n$ and whose other coordinates are 0.

Proposition 3.23. [∀ `finalLargeVector_properties`] *For every $n \in \mathbb{N}$ one has*

$$z_n \in B_X^{o(n)} \qquad \textit{and} \qquad \|z_n\| = 2^n.$$

Proof. By Proposition 3.21, $2^n\mathbf{1} \in B_{X_n}^{o(n)}$. Identifying X_n with the nth coordinate band of X, the same directed families witness $z_n \in B_X^{o(n)}$. Also $\|z_n\| = p_n(2^n\mathbf{1}) = 2^n$. □

Proof of Theorem 3.1. By Lemma 3.22, X is weakly Fatou.

Assume, towards a contradiction, that X admits an equivalent lattice norm σ with the Fatou property. Then there exist constants $c, C > 0$ such that

$$c^{-1}\|x\| \leqslant \sigma(x) \leqslant C\|x\| \qquad (x \in X).$$

Hence

$$B_X \subseteq C\, B_\sigma.$$

By Lemmas 3.2 and 3.3,

$$B_X^{o(n)} \subseteq C\, B_\sigma^{o(n)} = C\, B_\sigma \subseteq Cc\, B_X \qquad (n \geqslant 1).$$

Thus, every element of every iterated order adherence of B_X has $\|\cdot\|$-norm at most Cc. However, Proposition 3.23 gives $z_n \in B_X^{o(n)}$ and

$$\|z_n\| = 2^n \to \infty,$$

a contradiction. Therefore, X is not equivalent to any Fatou lattice norm. □

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